

# Lab 3 – Architectural Justification

## Campus Placement & Internship Pipeline (CPIP)

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### Architecture selection:

We chose Layered Architecture for the Campus Placement & Internship Pipeline (CPIP) System. The architecture separates the presentation, business/application and data responsibilities while keeping the placement workflow organized.

### Two scenario-related reasons:

1. CPIP has clear functional stages - student interaction, placement/application processing, eligibility and resume processing, offer/notification handling and persistent data - so separation into layers keeps these responsibilities modular & easier to maintain.
2. CPIP is a centralized placement workflow rather than a collection of independently deployable services. A layered structure supports the existing “profile -> eligibility -> application/interview -> offer” flow without introducing the operational complexity & distributed-service overhead associated with Microservices. Compared with Client-Server, it provides clearer separation between presentation, business processing & data management.

### Security advantage:

The layered structure places sensitive academic records, resume information, application results & offer documents behind the business/application layer rather than exposing the data layer directly to the user interface. The data layer can enforce the Lab 1 requirement (in the Lab 1 pdf) for AES-256 encryption of student academic records and offer documents at rest, while application-layer access control can regulate access to these records.

### Performance benefit:

Eligibility evaluation is kept within the business/application layer, avoiding unnecessary distributed-service communication for a core operation. This supports the Lab 1 performance requirement (in the Lab 1 pdf) that eligibility evaluation results be returned within 3 seconds for at least 95% of requests under normal operating conditions.

### Component diagram:

